

# Multiplicative renormalization of composite operators

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(Received 24 June 1975)

We study composite operators with the aid of dimensional regularization. Using Zimmermann "normal product algorithm," we obtain a simple formula for any normal product in renormalized field theory and discuss some of its properties.

## I. INTRODUCTION

The problem of defining renormalized composite operators in QFT has been treated by several authors using different approaches.<sup>1</sup> Among them, Zimmermann has developed a powerful formalism called the "normal product algorithm,"<sup>2</sup> in which composite operators are defined in Green's functions, and renormalized afterwards with the BPHZ method.<sup>3</sup>

With the development of the dimensional regularization method<sup>4</sup>—which as it is well-known, presents noteworthy properties—comes the possibility of introducing composite operators (CO) in the conventional renormalization approach.<sup>5</sup> They are defined using the number of dimensions ( $\nu$ ) as a regulator, and coincide with the CO introduced by Zimmermann when  $\nu = 4$ .

This new approach simplifies explicit calculations with CO and provides a new method to study different properties in QFT such as generalized Ward identities anomalies, etc. We present here an investigation in that direction. We establish (following Zimmermann's proof of the so-called "Zimmermann identities,"<sup>2</sup> but in the context of dimensional regularization) the possibility of obtaining any normal product  $N[\Theta(x)]$  as a linear combination of a finite number of regularized CO. Those operators share the quantum numbers of  $\Theta(x)$  and have a dimension that, at most, equals that of  $\Theta(x)$ .

The relation between normal products and regularized CO that we obtain, are related to Zimmermann identities between normal products of different degrees.<sup>2</sup> In our case the renormalization constants and the regularized Green's functions have poles at  $\nu = 4$ . The objects in Zimmermann's identities between normal products  $N_d[\Theta]$  with  $d \geq \dim \Theta$  are all finite in four dimensions. Actually, the expressions that we obtain correspond to a Zimmermann identity, in  $\nu$  dimensions, between a normal product  $N_{-1}[\Theta]$  and  $N_d[\Theta]$  ( $d \geq \dim \Theta$ ) sufficiently renormalized.

The above mentioned expression for a normal product is obtained in Sec. II. In Sec. III we discuss some properties of the new renormalization constants and state some concluding remarks.

## II. REGULARIZED AND RENORMALIZED OPERATORS

The aim of this section is to present a derivation of the normal product expansion for any CO in terms of  $\nu$ -dimensional regularized operators.

For the sake of clarity, we begin by considering a model of a scalar field with a  $g\phi^4$  coupling, but all our results will be valid for any other renormalizable quantum field theory. We use the dimensional regularization method,<sup>4</sup> which as is well known, shows remarkable advantages over the other previous methods. In particular, it was chosen due to its property of preserving symmetries and invariances. Nevertheless, it is worthwhile mentioning that all that follows might have been done with any other regularization method sharing those properties.

We start by considering the standard Lagrangian that includes all counterterms

$$\mathcal{L}(x) = \frac{1}{2}(Z_2 + z_2)(\partial\phi)^2 - \frac{1}{2}(M^2 + \delta M^2 + \delta m^2)\phi^2 - (g/4!)(Z_1 + z_1)\phi^4; \quad (2.1)$$

here,  $Z_2$ ,  $\delta M^2$ , and  $Z_1$  are computed from the one-particle irreducible (1PI) diagrams with two and four external legs, taken at zero external momenta. In other words, these counterterms eliminate all UV divergencies from the Green's functions (GF). In Eq. (2.1),  $z_2$ ,  $\delta m^2$ , and  $z_1$  are finite renormalizations.

The connected GF are then specified, in perturbation theory by the Gell-Mann–Low formula

$$\begin{aligned} G^{(N)}(x_1, x_2, \dots, x_N; M, g, \nu) \\ = \langle T\phi(x_1) \cdots \phi(x_N) \rangle^{\text{conn}} \\ = \langle 0 | T\phi^{(0)}(x_1) \cdots \phi^{(0)}(x_N) \exp[i \int \mathcal{L}_i(x) d^4x] | 0 \rangle_{\otimes}^{\text{conn}}. \end{aligned} \quad (2.2)$$

Here  $\phi^{(0)}(x)$  is the free scalar field,  $|0\rangle$  the free field vacuum state, and with the symbol  $\otimes$  we are indicating that we omit all vacuum–vacuum diagrams. The interaction Lagrangian  $\mathcal{L}_i(x)$  is

$$\mathcal{L}_i(x) = \mathcal{L}(x) - [\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}M^2\phi^2(x)]. \quad (2.3)$$

As it was noticed above, all preceeding expressions are considered in a  $\nu$ -dimensional space.

Now, these GF, order by order in perturbation theory, are meromorphic functions of  $\nu$  and its limit when  $\nu \rightarrow 4$  exists and is finite.<sup>4,6</sup> As it was stressed in the Introduction, an alternative way of obtaining GF has been given by Zimmermann<sup>2</sup> without using regulators (obviously working in a four-dimensional space). This

method gives

$$G^{(N)}(x_1, \dots, x_N; M, g, 4) = \text{PF} \langle 0 | T \phi^{(0)}(x_1) \cdots \phi^{(0)}(x_N) \rangle \times \exp[i \int \mathcal{L}_{\text{eff}}^{(0)}(x) d^4x] | 0 \rangle_{\otimes}^{\text{conn}}, \quad (2.4)$$

with

$$\mathcal{L}_{\text{eff}}^{(0)}(x) = \frac{1}{2}(z_2 - 1)(\partial\phi)^2 - (\delta m^2/2)\phi(x)^2 - (g/4!)(1 + z_1)\phi^4(x). \quad (2.5)$$

Here PF means that we must consider the BPHZ finite part diagram contributions computed at zero external momenta.

For our purposes, we must have the previous BPHZ formula (2.4) extended to a  $\nu$ -dimensional space. This is a straightforward generalization and reads

$$G^{(N)}(x_1, \dots, x_N; M, g, \nu) = \text{PF} \langle 0 | T \phi^{(0)}(x_1) \cdots \phi^{(0)}(x_N) \rangle \times \exp[i \int \mathcal{L}_{\text{eff}}^{(0)}(x) d^\nu x] | 0 \rangle_{\otimes}^{\text{conn}}. \quad (2.6)$$

Here  $z_1$ ,  $z_2$ , and  $\delta m^2$ , the finite counterterms of the  $\mathcal{L}_{\text{eff}}$  should be considered as functions of  $\nu$ . Obviously, expressions (2.2) and (2.6) are free from UV divergencies in  $\nu=4$ . But once we have taken the limit  $\nu \rightarrow 4$ , if in such renormalized GF, we make  $x_1 = x_2 = \dots = x_j$  ( $j=2, 3, \dots, N$ ), it is well known that the resulting expression diverges. This disease shows the necessity of giving renormalized expressions for the so-called composite operators—monomials of the basic field  $\phi(x)$  and its derivatives. As we have mentioned, there exists a definition of a GF with a CO, due to Zimmermann, that we will use as the starting point of our derivation. Zimmermann's expressions when written in  $\nu$  dimensions read

$$\langle T \mathcal{N}[\Theta(x)] \phi(x_1) \cdots \phi(x_N) \rangle = \text{PF} \langle 0 | \Theta^{(0)}(x) \phi^{(0)}(x_1) \cdots \phi^{(0)}(x_N) \rangle \times \exp[i \int \mathcal{L}_{\text{eff}}^{(0)}(x) d^\nu x] | 0 \rangle_{\otimes}^{\text{conn}}, \quad (2.7)$$

where  $\Theta^{(0)}(x)$  is obtained from the CO  $\Theta(x)$  by the substitution  $\phi(x) \rightarrow \phi^{(0)}(x)$ .

Let us point out that if we take  $x_1 = x_2 = \dots = x_j$  ( $j=2, 3, \dots, N$ ) in a renormalized GF maintained in  $\nu$  dimensions, the resulting expression remains mathematically well defined, provided  $\nu$  is sufficiently negative. The second step will be to derive an expression for a nonrenormalized CO, meromorphic function of the dimension  $\nu$ .<sup>7</sup>

We define the regularized CO in terms of its expression in momentum space

$$\Theta_{\{\mu\}}(x) = \prod_{i=1}^m \partial_{\mu_i} \phi(x), \quad (2.8)$$

$$\partial_{(\mu)_i} \equiv \partial_{\mu_{j_1}} \cdots \partial_{\mu_{j_m}(\mu_i)}; \partial_\mu = \frac{\partial}{\partial x^\mu}; \{\mu\} \equiv \{(\mu)_1, \dots, (\mu)_m\},$$

by the expression

$$\langle T \Theta_{\{\mu\}}(x) \phi(x_1) \cdots \phi(x_N) \rangle^{\text{conn}} = \int \prod_{i=1}^m \frac{d^p q_i}{(2\pi)^\nu} \prod_{j=1}^N \frac{d^p p_j}{(2\pi)^\nu}$$

$$\times \exp[-i(\sum_{j=1}^N p_j \cdot x_j + x \cdot \sum_{i=1}^m q_i)] P_{\{\mu\}} \times G^{(N+m)}(q_1, \dots, q_m, p_1, \dots, p_N; M, g, \nu), \quad (2.9)$$

where

$$P_{\{\mu\}} = \prod_{i=1}^m q_{i,(\mu)_i} (-i)^{\sum_{j=1}^m \mu_{j_i}}, \quad q_{i,(\mu)_i} = q_{i,\mu_j}, \dots, q_{i,\mu_{j_m}(\mu_i)}.$$

The perturbative expansion of the rhs of Eq. (2.9) contains diagrams with  $(m+N)$  legs;  $m$  of those legs have been identified forming the composite vertex and also the factor  $P_{\{\mu\}}$  has been added. Note that even the mentioned diagrams are different from those contributing to an ordinary GF; in the context of dimensional regularization, they have UV divergencies which, like for ordinary GF, appear as poles in  $\nu=4$ . [See for example Ref. 6.] It is clear that these poles are not eliminated by the Lagrangian counterterms, because they come from the fact that arguments have been identified in a renormalized GF in configuration space.

We shall now prove that the renormalized operator can be obtained from the following linear combination of regularized operators:

$$\langle T \mathcal{N}[\Theta_{\{\mu\}}] \phi(x_1) \cdots \phi(x_N) \rangle = \sum_{\Theta_{\{\rho\}}} (Z^{-1})_{\Theta_{\{\mu\}} \Theta_{\{\rho\}}}^{\Theta_{\{\rho\}}} \langle T \Theta_{\{\rho\}} \phi(x_1) \cdots \phi(x_N) \rangle, \quad (2.10)$$

where the sum  $\sum_{\Theta_{\{\rho\}}}$  extends over all operators  $\Theta_{\{\rho\}}$  with

$$D_{\Theta_{\{\rho\}}} = r + \sum_{j=1}^r r(j) < m + \sum_{j=1}^m m(j) = D_{\Theta_{\{\mu\}}}. \quad (2.11)$$

That is to say, the sum extends over all those CO  $\Theta_{\{\mu\}}$  which, in  $\nu=4$ , have a dimension—in mass units—less than or equal to the dimension of  $\Theta_{\{\mu\}}$ . We also notice that those operators must have the same transformation properties as  $\Theta_{\{\mu\}}$  under the symmetries of the theory.

On the other hand, the renormalization constants  $(Z^{-1})_{\Theta_{\{\mu\}} \Theta_{\{\rho\}}}^{\Theta_{\{\rho\}}}$  are invariant tensors, coordinate-independent, which in general have poles, order by order in perturbation theory, at  $\nu=4$ . They are responsible for the renormalization of the CO.

One can derive Eq. (2.10) starting with a Zimmermann identity between nonlocal product of operators, in  $\nu$  dimensions, and then taking the limit when all points become identical. However, we sketch another demonstration in which it is not necessary to make use of point splitting because one deals with  $\nu$ -regularized expressions from the beginning. This derivation is analogous to that corresponding to Zimmermann identities between operators of different subtraction degree.<sup>2</sup>

Let us consider a diagram  $\Delta$  which corresponds to the perturbative expansion of the GF, as given by the lhs of Eq. (2.10). In momentum space, this diagram has  $N$  external legs corresponding to the field  $\phi$  and a composite to vertex  $V_0$  corresponding to the CO  $\Theta_{\{\mu\}}$ . This diagram  $\Delta$  can be obtained from a connected diagram  $\Gamma$ , belonging to the connected GF  $G^{(N+m)}$ , when we collapse  $m$  of its legs, though they form the composite vertex  $V_0$ . (We have also to multiply the resulting expression by the factor  $P_{\{\mu\}}$ .)

In the BPHZ scheme, the renormalized integrand corresponding to  $\Delta$  is

$$R_\Delta = S_\Delta \sum_{\mathbf{U} \in U(\Delta)} \prod_{\gamma \in \mathbf{U}} (-\ell^\gamma S_\gamma) S_{\mathbf{U}} I_\Delta. \quad (2.12)$$

Here,  $I_\Delta$  is the unsubtracted integrand and  $S_{\mathbf{U}}$ ,  $S_\gamma$ , and  $S_\Delta$  are the substitution operators (see Ref. 8).  $U(\Delta)$  is the set of all  $\Delta$ -forests and the  $\ell^\gamma$  are the Taylor operators at order  $d(\gamma)$  in the external momenta of  $\gamma$ . Finally, the degree function  $d(\gamma)$  is given by

$$d(\gamma) = \begin{cases} 4 - r(\gamma), & \text{if } V_0 \notin \gamma, \\ D_{\Theta_{\{\mu\}}} - r(\gamma), & \text{if } V_0 \in \gamma. \end{cases} \quad (2.13)$$

[We take  $\ell^0 \equiv 0$  for  $d(\gamma) < 0$ .] In this last equation,  $r(\gamma)$  denotes the number of external lines of  $\gamma$ .

The renormalized integrand corresponding to the above mentioned diagram  $\Gamma$  is

$$R_\Gamma = S_\Gamma \sum_{\mathbf{U} \in U(\Gamma)} \prod_{\gamma \in \mathbf{U}} (-\ell^\gamma S_\gamma) S_{\mathbf{U}} I_\Gamma. \quad (2.14)$$

The renormalized integrand  $R_\Delta$  can be expressed as

$$R_\Delta = S_\Delta P_{\{\mu\}} R_\Gamma + X, \quad (2.15)$$

where  $X$  can be written as<sup>2</sup>

$$X = -S_\Delta \sum_{\tau \in T_\alpha} \sum_{\{\mu\}} [(-i)^{\sum_{j=1}^r r(j)} / \prod_{j=1}^r r(j)!] \\ \times R_{\Delta/\tau\{\mu\}} [\partial_{p_1}^{(p)} \dots \partial_{p_r}^{(p)}] R_\tau' \Big|_{p_j=0}. \quad (2.16)$$

Here  $\tau$  is the smallest element belonging to  $U$ , containing  $V_0$ ,  $T_\alpha$  is the set of all proper subdiagrams  $\tau$  of  $\Delta$ ,  $\mu_\tau$  is the set of all  $\Delta$ -forests  $\mathbf{U}_1$  which display the property of each  $\gamma_1 \in \mathbf{U}_1$  satisfying

$$\tau \subset \gamma_1 \ (\tau \neq \gamma) \text{ or } \gamma \cap \gamma_1 = \emptyset,$$

and  $U(\tau)$  is the set of all  $\tau$ -forest not containing  $\tau$ . The sum extends to

$$\sum_{j=1}^r r(j) \leq D_{\Theta_{\{\mu\}}} - r; \quad r \equiv r(\tau) \quad (2.17)$$

and

$$R_{\Delta/\tau\{\mu\}} = \sum_{\mathbf{U}_1 \in \mu_\tau} \prod_{\gamma_1 \in \mathbf{U}_1} (-\ell^{\gamma_1} S_{\gamma_1}) \\ \times \prod_{j=1}^r (-i l_{a_i b_i \sigma_i}^\mu) u_{ij} S_{\mathbf{U}_1} I_{\Delta/\tau}, \quad (2.18)$$

$$R_\tau' = \sum_{\mathbf{U}_2 \in U(\tau)} \prod_{\gamma_2 \in \mathbf{U}_2} (-\ell^{\gamma_2} S_{\gamma_2}) S_{\mathbf{U}_2} I_\tau, \quad (2.19)$$

where the  $l_{a_i b_i \sigma_i}$  are the momenta arriving to  $\tau$ .

Here it may be noticed that the integrand given by Eq. (2.19) is not renormalized. Actually, forests containing  $\tau$  are not subtracted and therefore, when internal momenta integrations are performed, poles will appear at  $\nu=4$ . The integration of Eq. (2.16) over the internal momenta, followed by the sum over all  $\Delta$  and the Fourier transformation with respect to  $p, p_1, \dots, p_N$

lead to

$$\langle T\Theta_{\{\mu\}}(x) \phi(x_1) \dots \phi(x_N) \rangle \\ = \sum_{\Theta_{\{\mu\}}} Z_{\Theta_{\{\mu\}}}^{\Theta_{\{\mu\}}} \langle TN[\Theta_{\{\mu\}}] \phi(x_1) \dots \phi(x_N) \rangle, \quad (2.20)$$

where

$$Z_{\Theta_{\{\mu\}}}^{\Theta_{\{\mu\}}} = \delta_{\Theta_{\{\mu\}}}^{\Theta_{\{\mu\}}} + (-i)^{\sum_{j=1}^r r(j)} / \prod_{j=1}^r r(j)! \\ \times \sum_{\tau \in F_\tau} \partial_{q_1}^{(p)} \dots \partial_{q_r}^{(p)} \langle TN^\tau[\Theta_{\{\mu\}}] \tilde{\phi}(q_1) \dots \tilde{\phi}(q_r) \rangle_{a, i\tau=0}^{1PI}. \quad (2.21)$$

Here,  $F_\tau$  is the set of all nontrivial diagrams contributing to the proper part and  $N^\tau$  means that the integrand contributing to this expression are not the renormalized ones but those given by Eq. (2.19). The sum in Eq. (2.20) extends to all  $\Theta_{\{\mu\}}$  with the dimension  $D_{\Theta_{\{\mu\}}} \leq D_{\Theta_{\{\mu\}}}$ .

The renormalization matrix  $Z$  is triangular and at zero order in perturbation theory,  $\det Z = 1$ . Therefore, we can invert Eq. (2.20) and thus obtain Eq. (2.10).

In the case of oversubtracted normal products,  $N_\alpha[\Theta]$ , we have similarly

$$\langle TN_\alpha[\Theta_{\{\mu\}}(x)] \phi(x_1) \dots \phi(x_N) \rangle \\ = \sum_{\Theta_{\{\mu\}}} (Z_\alpha^{-1})_{\Theta_{\{\mu\}}}^{\Theta_{\{\mu\}}} \langle T\Theta_{\{\mu\}} \phi(x_1) \dots \phi(x_N) \rangle, \quad (2.22)$$

where the sum  $\sum_{\Theta_{\{\mu\}}}$  extends to all CO with dimension  $D_{\Theta_{\{\mu\}}} \leq \alpha$ .

It is obvious that Eqs. (2.20) show the case in which a composite operator is renormalizable in a multiplicatively way: when there are no operators with the same quantum numbers and dimension less than or equal to it.

### III. PROPERTIES OF THE RENORMALIZATION CONSTANTS

In this section we will study some properties of the renormalization constants  $Z_{\Theta_{\{\mu\}}}^{\Theta_{\{\mu\}}}$  from the normal product properties.

The constants  $Z_{\Theta_{\{\mu\}}}^{\Theta_{\{\mu\}}}$  may either be determined directly from Eq. (2.21), or more easily, by application of the normalization condition of the normal products.

For example, in the case of the operator  $\phi^4(x)$

$$Z_{\phi^4}^{\phi^4} = \frac{1}{2} \langle T\phi^4(0) \tilde{\phi}(0) \tilde{\phi}(0) \rangle^{1PI}, \\ Z_{\phi^4}^{\phi^4} = \frac{1}{4} \langle T\phi^4(0) \tilde{\phi}(0) \tilde{\phi}(0) \tilde{\phi}(0) \tilde{\phi}(0) \rangle \\ - (1/4!) Z_{\phi^4}^{\phi^4} / Z_{\phi^2}^{\phi^2} \langle T\phi^2(0) \tilde{\phi}(0) \tilde{\phi}(0) \tilde{\phi}(0) \tilde{\phi}(0) \rangle^{1PI}.$$

One also gets expressions for  $Z_{\phi^2}^{\phi^2}$ —working with  $N[\phi^2]$

$$Z_{\phi^2}^{\phi^2} = \frac{1}{2} \langle T\phi^2(0) \tilde{\phi}(0) \tilde{\phi}(0) \rangle^{1PI}$$

and similarly for the remaining constants which renormalize  $\phi^4$  ( $Z_{\phi^4}^{\phi^4}$  and  $Z_{\phi^4}^{(\partial^2 \phi^4)}$ ). From the linearity of normal products and the differentiation formula, one

derives

$$\begin{aligned}(Z^{-1})_{\theta_1 \theta_2}^{\theta} &= (Z^{-1})_{\theta_1}^{\theta} + (Z^{-1})_{\theta_2}^{\theta}, \\ (Z^{-1})_{\theta_2}^{\theta_1} &= (Z^{-1})_{\theta_2}^{\theta_1}.\end{aligned}\quad (3.1)$$

Finally, we shall obtain relations between CO renormalization constants and the renormalization constants appearing in the Lagrangian (2.1). In order to do this, we shall study the equation of motion satisfied by connected GF. Making use of the generalized Wick's theorem<sup>10</sup>

$$\begin{aligned}\langle 0 | T \phi^{(0)}(x_1) \cdots \phi^{(0)}(x_N) \phi^{(0)}(x) \exp[i \int \mathcal{L}_I^{(0)}(z) d^4 z] | 0 \rangle_{\otimes}^{\text{conn}} \\ = i \int d^4 w \langle 0 | T \phi^{(0)}(x_1) \cdots \phi^{(0)}(x_N) \overline{\phi^{(0)}(x) \mathcal{L}^{(0)}(w)} \\ \times \exp[i \int \mathcal{L}_I^{(0)}(z) d^4 z] | 0 \rangle_{\otimes}^{\text{conn}} + i \delta_{N1} \Delta_F(x - x_1).\end{aligned}\quad (3.2)$$

With the use of  $\mathcal{L}_I$  given by Eqs. (2.4)–(2.1) we obtain

$$\begin{aligned}G^{(N+1)}(x, x_1, x_2, \dots, x_N; M, g, \nu) &= i \delta_{N1} \Delta_F(x - x_1) \\ &+ \int d^4 w \Delta(x - w) \langle 0 | T \{ (Z_2 + z_2 - 1) \square \phi(w) + (\delta m^2 + \delta M^2) \phi(w) \\ &+ (g/3!) (Z_1 + z_1) \phi^3(w) \} \phi(x_1) \cdots \phi(x_N) \\ &\exp[i \int \mathcal{L}_I^{(0)}(z) d^4 z] | 0 \rangle_{\otimes}^{\text{conn}}\end{aligned}\quad (3.3)$$

where

$$\langle 0 | T \phi^{(0)}(x) \phi^{(0)}(y) | 0 \rangle = i \Delta_F(x - y) = \overline{\phi(x) \phi(y)}.$$

Applying the Klein–Gordon differential operator, we finally have

$$\begin{aligned}(\square_x + m^2) \langle T \phi(x) \phi(x_1) \cdots \phi(x_N) \rangle &= - \langle T \{ (\delta M^2 + \delta m^2) \phi(x) \\ &+ (Z_2 + z_2 - 1) \square \phi(x) + (g/3!) (Z_1 + z_1) \phi^3(x) \} \phi(x_1) \cdots \phi(x_N) \rangle \\ &- i \delta_{N1} \delta(x - x_1).\end{aligned}\quad (3.4)$$

Subtracting the equation of motion derived with the aid of the BPHZ algorithm from Eq. (3.4), we obtain

$$Z_{\phi^3}^{\phi} = Z_1, \quad g Z_{\phi^3}^{\phi} = 3! \delta M^2, \quad g Z_{\phi^3}^{\square \phi} = 3! (Z_2 - 1). \quad (3.5)$$

## ACKNOWLEDGMENT

We are indebted to C. Garcia Canal for a critical reading of the manuscript.

\*Fellow of the Consejo Nacional de Investigaciones Científicas y Técnicas, Argentina.

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<sup>1</sup>H. Epstein and V. Glaser, in *Statistical Mechanics and QFT*, Les Houches (Gordon and Breach, New York, 1970); K. Wilson (unpublished) quoted in Ref. 8; W. Zimmermann, Refs. 2 and 8.

<sup>2</sup>W. Zimmermann, in *Brandeis Lectures on Elementary Particles and QFT*, edited by S. Deser, M. Grisaru, and H. Pendleton (MIT Press, Cambridge, Mass., 1971), Vol. I. See also Ref. 8.

<sup>3</sup>N.N. Bogoliubov and O.S. Parasiuk, *Acta Math.* **97**, 227 (1957); K. Hepp, *Commun. Math. Phys.* **2**, 301 (1966); W. Zimmermann, *Commun. Math. Phys.* **15**, 208 (1969).

<sup>4</sup>C.G. Bollini and J.J. Giambiagi, *Phys. Lett. B* **40**, 566 (1972); *Nuovo Cimento B* **12**, 20 (1972); G. 't Hooft and M. Veltman, *Nucl. Phys. B* **44**, 189 (1972).

<sup>5</sup>See for example J.D. Bjorken and S.D. Drell, *Relativistic Quantum Fields* (McGraw-Hill, New York, 1965).

<sup>6</sup>H. de Vega and F. Schaposnik, *J. of Math. Phys.* **15**, 1998 (1974).

<sup>7</sup>Some particular cases of CO in dimensional regularization have already been considered in Ref. 9.

<sup>8</sup>W. Zimmermann, *Ann. Phys.* **77**, 536 (1973).

<sup>9</sup>C.G. Bollini and J.J. Giambiagi, *Acta Phys. Austriaca* **38**, 210 (1972); H. Girotti, *Nucl. Phys. B* **64**, 244 (1973); H. de Vega and H. Girotti, *Nucl. Phys. B* **79**, 77 (1974).

<sup>10</sup>N.N. Bogoliubov and D.V. Shirkov, *Introduction to the Theory of Quantized Fields* (Interscience, New York, 1959).